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12. mean $(\bar{x}) = \frac{\sum_{i=1}^{n} x_i}{n}$

$= \frac{34.61+34.57+34.40+34.63+34.63+34.51+34.49+34.61+34.52+34.55+34.58+34.53+34.44+34.48+34.40}{15}$

$= 34.53$

Standard deviation $= \sqrt{\sigma^2}$

Variance $(\sigma^2) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$

$= \frac{(34.61-34.53)^2 + (34.57-34.53)^2 \ldots + (34.40-34.53)^2}{14}$

$= \frac{0.0064+0.0016+0.0169+0.01+0.01+0.0004+0.0016+0.0064+0.0001+0.0004+0.0025+0+0.0081+0.0025+0.0169}{14}$

$= 0.0059$

$= \frac{0.0838}{15} = 0.00558$

Standard deviation $= \sqrt{\sigma^2} = \sqrt{0.0059} = 0.0768$

$= \sqrt{0.00558} = 0.07469$

$= 0.0747$

Teacher's Signature : _____

13. KNN

Given, k = 3

Students	X (Academic Excellence)	Y (Activities)	Z (Classification)
S_1	8	6	Outstanding
S_2	5	6	Good
S_3	7	3	Good
S_4	6	9	Outstanding
S_5	5	7	?

Find Euclidean distance of new inputs (5, 7) with rest other data points.

$(S_5, S_1) = \sqrt{(5-8)^2 + (7-6)^2} = 3.16$

$(S_5, S_2) = \sqrt{(5-5)^2 + (7-6)^2} = 1$

$(S_5, S_3) = \sqrt{(5-7)^2 + (7-3)^2} = 4.47$

$(S_5, S_4) = \sqrt{(5-6)^2 + (7-9)^2} = 2.23$

Since, k = 3 Pick the least 3 distances and their corresponding classes.

2. $(S_5, S_1) = 3.16$ Outstanding
   $(S_5, S_2) = 1$ Good
   $(S_5, S_4) = 2.23$ Outstanding

majority
Outstanding
So, (5, 7) : Outstanding

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14. Naive Bayes

Find Class of (Swim: slow, Fly: Rarely, Crawl: No)

P(class: fish) = 3/12 P(class=Bird) = 4/12

P(class = Animal) = 5/12

~~Swim~~ ~~Yes~~ No

~~Fast~~

~~slow~~

~~No~~

~~Fly~~ ~~Yes~~ ~~No~~

~~Long~~

~~short~~

~~Rarely~~

~~No~~

Swim	Fish	Animal	Bird
Fast	1/3	2/5	1/4
slow	2/3	2/5	0/4
No	0/3	1/5	3/4

Crawl	Fish	Animal	Bird
Yes	1/3	2/5	0/4
No	2/3	3/5	4/4

Fly	Fish	Animal	Bird
Long	0/3	0/5	1/4
short	0/3	0/5	2/4
Rarely	0/3	1/5	0/4
No	3/3	4/5	1/4

$V_{NB}(fish) = P(class=fish)\, P(swim:slow)\, P(fly=rarely)\, P(Crawl=No)$

$= 3/12 \cdot 2/3 \cdot 0/3 \cdot 2/3 = 0$

Teacher's Signature :

$V_{NB}(Animal) = P(class=Animal) \cdot P(swim=slow) \cdot P(fly=rarely) \cdot P(crawl=No)$

$= 5/12 \cdot 2/5 \cdot 1/5 \cdot 3/5 = 0.02$

$V_{NB}(Bird) = P(class=Bird) \cdot P(swim=slow) \cdot P(fly=rarely) \cdot P(crawl=No)$

$= 4/12 \cdot 0/4 \cdot 0/4 \cdot 4/4 = 0$

By using Laplace estimator as 1.

Solution

Class = Animal

$V_{NB}(fish) = 3/12 \cdot 3/6 \cdot 1/6 \cdot 3/6 = 0.0104$

$V_{NB}(Animal) = 5/12 \cdot 3/8 \cdot 2/8 \cdot 4/8 = 0.019$

$V_{NB}(Bird) = 4/12 \cdot 1/7 \cdot 1/7 \cdot 5/7 = 0.004$

Normalization

$V_{NB}(fish) = \frac{0.0104}{0.0104 + 0.019 + 0.004} = 0.311$

$V_{NB}(Animal) = 0.568$

$V_{NB}(Bird) = 0.119$

$V_{NB}(fish) + V_{NB}(Animal) + V_{NB}(Bird) = 0.998 \approx 1$

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15. (a) Entropy (S): $\sum_{i=1}^{c} -P_i \log_2 (P_i)$

no: of positive classification = 3
no: of negative classification = 3
Total no: of instances = 6

Entropy (S): $-\left(\frac{3}{6}\right)\log_2\left(\frac{3}{6}\right) - \left(\frac{3}{6}\right)\log_2\left(\frac{3}{6}\right)$

$= 0.5 + 0.5 = 1$

(b) Information Gain (S, A)

$= \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$

OR

$= \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \text{Entropy}(S_v)$

In a_2

Entropy (a_2 = true) $= -\frac{4}{6}\log_2\frac{4}{6} = 0.389$

Entropy (a_2 = false) $= -\frac{2}{6}\log_2\frac{2}{6} = 0.528$

Information gain = $1 - (0.389 + 0.528)$

$= 0.083$

Teacher's Signature :

19. Total no: of patients : 10000

Healthy = 9000

Sick : 1000

Sick Test

Positive

Sick people

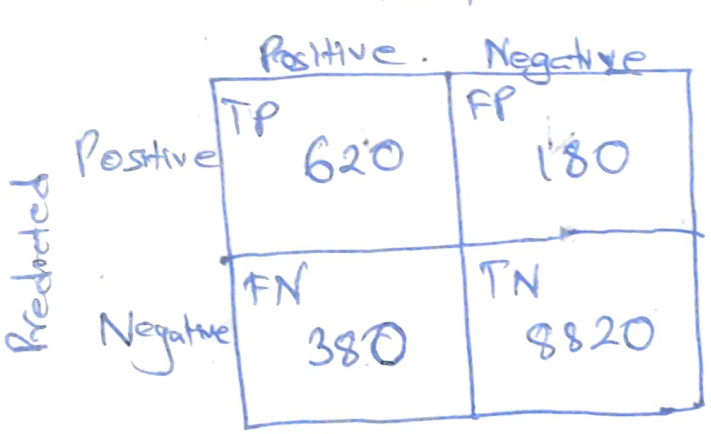
positive = 620

negative = 380

Health people

positive = 180

negative = 8820



$$\text{Recall} = \frac{TP}{TP + FN} = \frac{620}{620 + 380} = \underline{0.62}$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{620}{620 + 180} = \underline{0.775}$$